

RFTG



(~1996)

RFTG-m



(~1998)

RFG-M



(~2000)

No GPS,
1PPS or
RS422

RFTG-u



(~2000)

No Rb

RFTGm-II



(~2001)

RFTG



RFTG-m



Must exist; cannot find any picture

RFG-M



RFTG-u



Has no combiner/splitter

RFTGm-II



Members of the RFTG / RFG family

The nomenclature follows the nameplates on the devices.

Model	RFTG		RFTG-m		RFG-M (without the 'T') No GPS		RFTG-u Only as dual XO, no Rb		RFTGm-II	
Introduction year (approx.)	1996, three versions		1998		2000		2000		2000-2001	
Oscillator units (part # if known)	Rb (=REF0): KS24019 L102/L102A XO (=REF1): KS24019 L104/ L104A/L104B		RFTG-m-RB (1) KS24019 L105A RFTG-m-XO KS24019 L106A, L106B, L106C (also seen 2xRb units)		RFG-M-RB WP-92065 L11 RFG-M-XO WP-92066 L10		RFTG-u REF0 Z3812A RFTG-u REF1 Z3811A Pair + harness: Z3810AS		RFTGm-II-Rb KS-24019 L108 RFTGm-II-XO KS-24019 L109	
	Rb (left)	XO (right)	Rb	XO	Rb	XO	XO (REF0)	XO (REF1)	Rb	XO
15 MHz out (sine)	Only via combiner		Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
10 MHz out (square)	Yes		Yes		Yes (2x)		Yes		Yes (2x)	
10 MHz Ref In				Yes		Yes				Yes
1PPS (individual units)	No	No	Yes	Yes	No (?)	No (?)	Yes	Yes	Yes	Yes
1PPS (combined)	Yes		Yes		No		No		Yes	
Rb type	?		EFRATOM LPRO-101 (1)		?		No Rb		Symmetricom/Datum LPRO (3)	
OCXO type	?		EFRATOM 105243-001 (1)		?		Milliren (MTI) 260-0624-C (5.000MHz) (2)		EFRATOM 105243-003 (3)	
GPS receiver	In unit		In XO module (can discipline both) (Motorola Oncore)		No GPS		In REF1 module [likely Motorola Oncore]		In XO module (can discipline both) [likely Motorola Oncore]	
Meets Lucent spec.	KS24019		KS24019		?		KS24361		KS24019	
Physical controls	Manual switch for active unit		n/a		n/a		Manual switch for output level		n/a	
Notes	Purpose: Series II AMPS and CDMA system						"Built by HP/ Symmetricom" (16)			

Members of the RFTG family

Model	RFTG	RFTG-m or RFTGm-II	RFG-M (without the 'T')	RFTG-u
Fill ports & wire harness		Rudi's Electronics Lab [36]		
Power port	DSUB-15 (!)	P1', DSUB-9 male +24VDC, Pinout (26) (36)	P1', DSUB-9 male +24VDC	'P1', DSUB-9 male. +24VDC Pinout (14) ; between 18 and 36 Volts (15)
ALARM port	No such port	'J3', DSUB-9 female. Pinout and description, Confirmation –m and m-II are the same (26) (36)	'J3', DSUB-9 female. [No pinout found]	'J3', DSUB-9 female Opto isolator outputs. Pinouts (24) , meaning unknown
INTERFACE port (inter-unit)	[Not ports in the front, but via the frame, somehow the units must be connected]	'J5' DSUB-9 female; Description (9) Cable wiring and pinout (4) (12) ; GPS message routing (12) ; Cable wiring (5) , pinout likely same as (4) ; Full pinout, confirms it is same as –m (8) (26) (36)	DSUB-9 female [No pinout found]	'J5' DSUB-15 female (three-row); Pinout and measurement (13) ; Pinout (incomplete) (14) ; See also (2) ; Pinout & extensive description (15) ; Also has serial port, messages analysed by (14) ; Pinout and description (25) supplemented by (24) , Extensive description (27)
RS422/1PPS port	No separate port per unit, but RS422 and 1PPS are available via the combined 15-pin sockets	'J6' DSUB-9 female Pinout (1) (36) Confirmation –m and m-II are the same (26)	No such port	'J6' DSUB-9 female Pinout & extensive description (15) ; Details of RS422 talker messages (24) ; "speak only when spoken to" (25)
DIAGNOSTIC port	No such port	No such port	No such port	'J8' DSUB-9 female. Pinout and description, SCPI, 'same as J6 without PPS' (15) For commands and monitor status (source)
Connector panel / frame	In between the two units	-m: [Have not seen it, but probably exists?] m-II: yes, wide frame	Wide frame	[Don't think this exists]
Signal combiner/splitter (for 15 MHz)	6 outputs (TNC)	m-II: 2 inputs, 6 outputs (TNC)	2 inputs, 6 outputs (TNC)	No signal combiner/splitter
Signal combiner (for RS422 and 1PPS)	J7, J8, J9 (on frame, each DSUB-15). Each has 1 PPS, alarm and status signals. Uses cable with two DB9 RS422 connectors. (11)	J7, J8, J9 (on frame, each DSUB-15). Each has 1 PPS, alarm and status signals. Cabling (6, 7) (36)	No such port,	No signal combiner/splitter
Schematics	[Not found]	[Not found]	Reverse-engineered by Bud (1, 2)	[Not found]
Publicly available documentation	Reference Frequency and Timing Generator User Documentation Issue 1.0 November 1, 1996 (11) ; Lucent AUTOPLEX Cellular Telecommunications Systems System 1000 (29)	Short note (26)	[Not found]	[Not found]

Members of the RFTG family

Model	RFTG	RFTG-m RFTGm-II	RFG-M (without the 'T')	RFTG-u
RFTG 'Talker' mode with unit status, alarms, GPS time. Works via simple serial monitor (e.g. PuTTY on Windows or SerialTools by Kok Chen W7AY on MacOS).	On J7, J8, J9 (for both Rb and XO)	On Rb J6 and XO J6 Also on J7, J8, J9 on frame (for both Rb and XO)	No (lacks RS422 bus)	On REF0 J6 and REF1 J6. Serial monitor (14)
GPS 'Talker' mode with Motorola Oncore messages every ~5 seconds. @@ Ea (position), @@En (time), @@Bb (visible satellites) and @@Bo (UTC offset)	[Don't think so as there is no interface bus]	Yes, on XO J5 pin 2 (of 9 pins)	No (lacks GPS as well as RS422 bus)	Yes, on REF1 J5 pin1 (of 15 pins)
Command set for interactive control and communication (voltages, GPS data, etc.)	[Proprietary binary command set]	[Proprietary binary command set]	No (lacks RS422 bus)	'J8' DIAGNOSTIC port: Supports SCPI-like HP command set (16)
Software	HP RFTG utility ("RFTG Diagnostic Software package" (use 27 , not 28). (Use Windows XP or older OS). Communicates with both Rb and XO at the same time, when using two serial interfaces.	HP RFTG utility ("RFTG Diagnostic Software package" (use 27 , not 28). (Use Windows XP or older OS). Communicates with both Rb and XO at the same time, when using two serial interfaces. Mark S. Sims Lady Heather v6 Beta (20). Can run simultaneous instances for Rb and XO. Use the J6 RS422/1PPS port Works on Windows XP and Windows 11. Does not speak the "HP SCPI command set" so does not work with Symmetricom's SATSTAT software or the John MacConnell GPS.EXE utility (which looks for :ptim:tcd:cont 0)	No (lacks RS422 bus)	Mark S. Sims Lady Heather v5.01 regular edition (18) Symmetricom SATSTAT software (running in Windows XP), from 'J8' DIAGNOSTIC port) (2) (16) (30) HP GPS Control Program (running in Windows XP), from 'J8' DIAGNOSTIC port) (2) (16) (30) GPSCon software under Debian 7 and Wine (2) See also John MacConnell GPS.EXE and GPS-dos2.exe utilities (30)

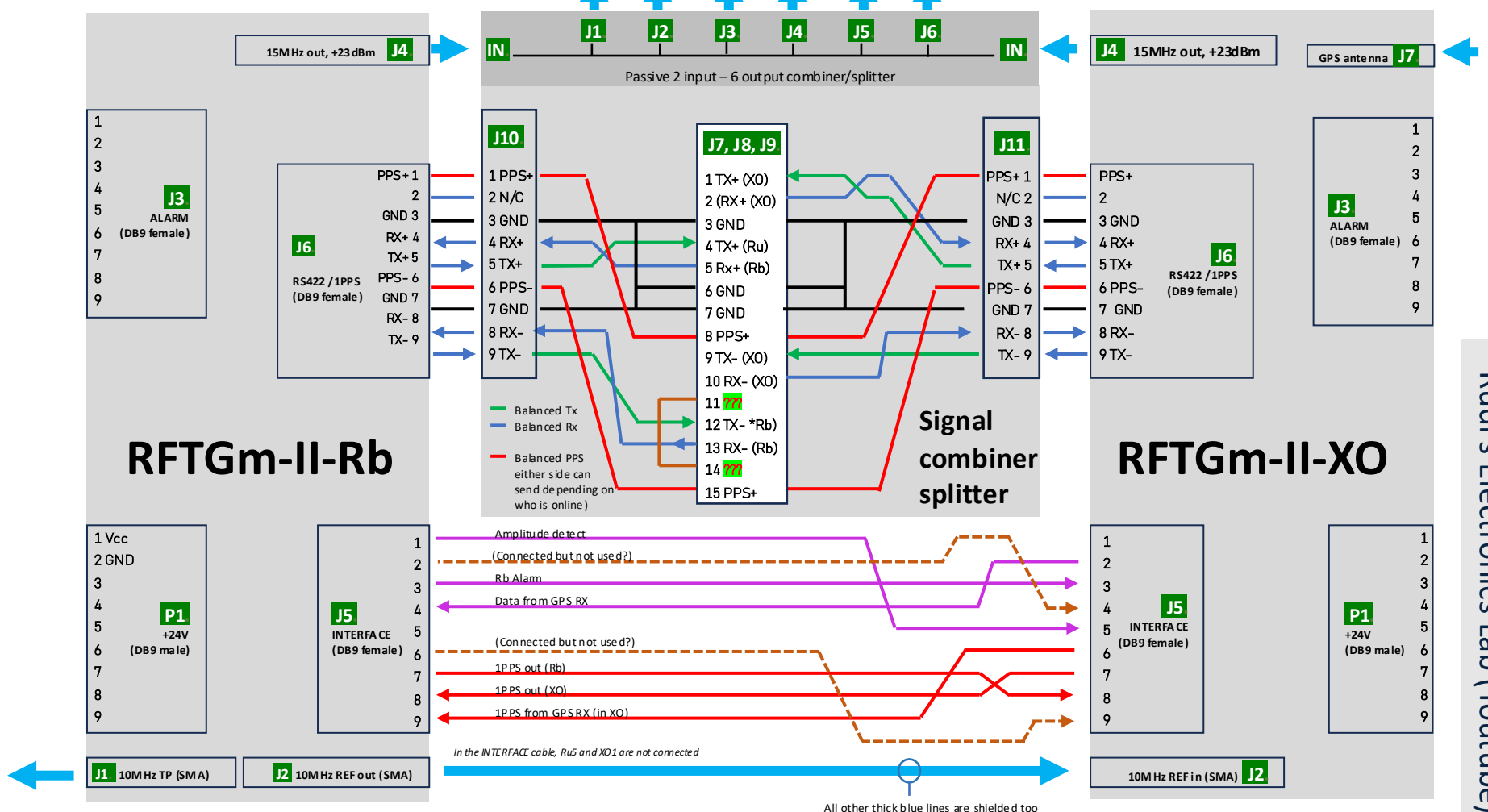
Sources

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- (3) Perdrix (David C. Partridge) (via EEVBLOG). <https://www.eevblog.com/forum/testgear/ks24019-rftgm-ii-rb-and-rftgm-ii-xo-teardown-10mhz-output-and-other-mods>
- (4) Pozdrawiam Wojtek SP7WK <http://bee.mif.pg.gda.pl/ciasteczki/kowypotwor/Lucent/RFTGm%20Interface%20Cable.pdf>
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- (6) Rodger Adams (via time-nuts). https://febo.com/pipermail/time-nuts_lists.febo.com/2020-July/100832.html
- (7) Unknown, 'Rodger'? (via time-nuts). https://febo.com/pipermail/time-nuts_lists.febo.com/attachments/20200714/d6b3aadd/attachment.pdf
- (8) ziggy9 (via time-nuts). https://febo.com/pipermail/time-nuts_lists.febo.com/2018-November/094691.html. Same post here: <https://www.mail-archive.com/time-nuts@lists.febo.com/msg01561.html>
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- (10) Pozdrawiam Wojtek SP7WK <http://bee.mif.pg.gda.pl/ciasteczki/kowypotwor/Lucent/RFTG-m%20RS422%20Intf.pdf>
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- (12) Pozdrawiam Wojtek SP7WK <http://bee.mif.pg.gda.pl/ciasteczki/kowypotwor/Lucent/RFTGm%20Message%20Routing.pdf>
- (13) Brooke Clarke <https://www.prc68.com/l/KS-24361.html>
- (14) Dan Watson (Sync Channel Blog) <https://syncchannel.blogspot.com/2015/08/standalone-operation-of-lucent-ks-24361.html>
- (15) Bob Camp (via time-nuts). <https://www.febo.com/pipermail/time-nuts/2015-August/093346.html>
- (16) Stewart Cobb (via time-nuts). <https://www.febo.com/pipermail/time-nuts/2014-October/087274.html>
- (17) Gerhard W. Hoffmann. <http://www.hoffmann-hochfrequenz.de/downloads/DoubDist.pdf>
- (18) Mark S. Sims e5fx. <http://www.ke5fx.com/heather/readme.htm>
- (19) Dan Watson (Sync Channel Blog) <https://syncchannel.blogspot.com/2015/08/adding-10mhz-output-to-ks-24361-ref-1.html>
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- (21) Tom Knox (via time-nuts). <https://lists.febo.com/empathy/thread/U62R2ZJPU36CRJSYSRJ5HU4Z6VAUFLT>
- (22) Skip Withrow. <https://telcodata.us/~myself/Lucent-RFTGm-Modification.pdf>
- (23) Charles S. Osborne k4cso (via time-nuts). <https://www.febo.com/pipermail/time-nuts/2008-April/030680.html>
- (24) Bob Camp (o-nyc.bodosom.net retrieved via wayback machine) <https://web.archive.org/web/20210227072243/http://do-nyc.bodosom.net/>
- (25) Stewart Cobb (o-nyc.bodosom.net retrieved via wayback machine) <https://web.archive.org/web/20210227072243/http://do-nyc.bodosom.net/>
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- (30). K8CO. <https://www.realhamradio.com/gps-satstat-software.htm>
- (31) Paul Wade W1GH. https://www.w1ghz.org/small_proj/Simple_Frequency_Doublers.pdf
- (32) Nigel GM8PZR https://www.w1ghz.org/small_proj/Simple_Frequency_Doublers.pdf
- (33) Motocoder. Reply #142 on <https://www.eevblog.com/forum/testgear/economical-option-for-precision-frequency-reference/?all>
- (34) Tom Knox. <https://list.mlfhs.org.uk/empathy/thread/U62R2ZJPU36CRJSYSRJ5HU4Z6VAUFLT>
- (35) John C. Roos, K6IQL. https://www.arl.org/files/file/OEX%20Binaries/11_March/3x11_Roos.zip
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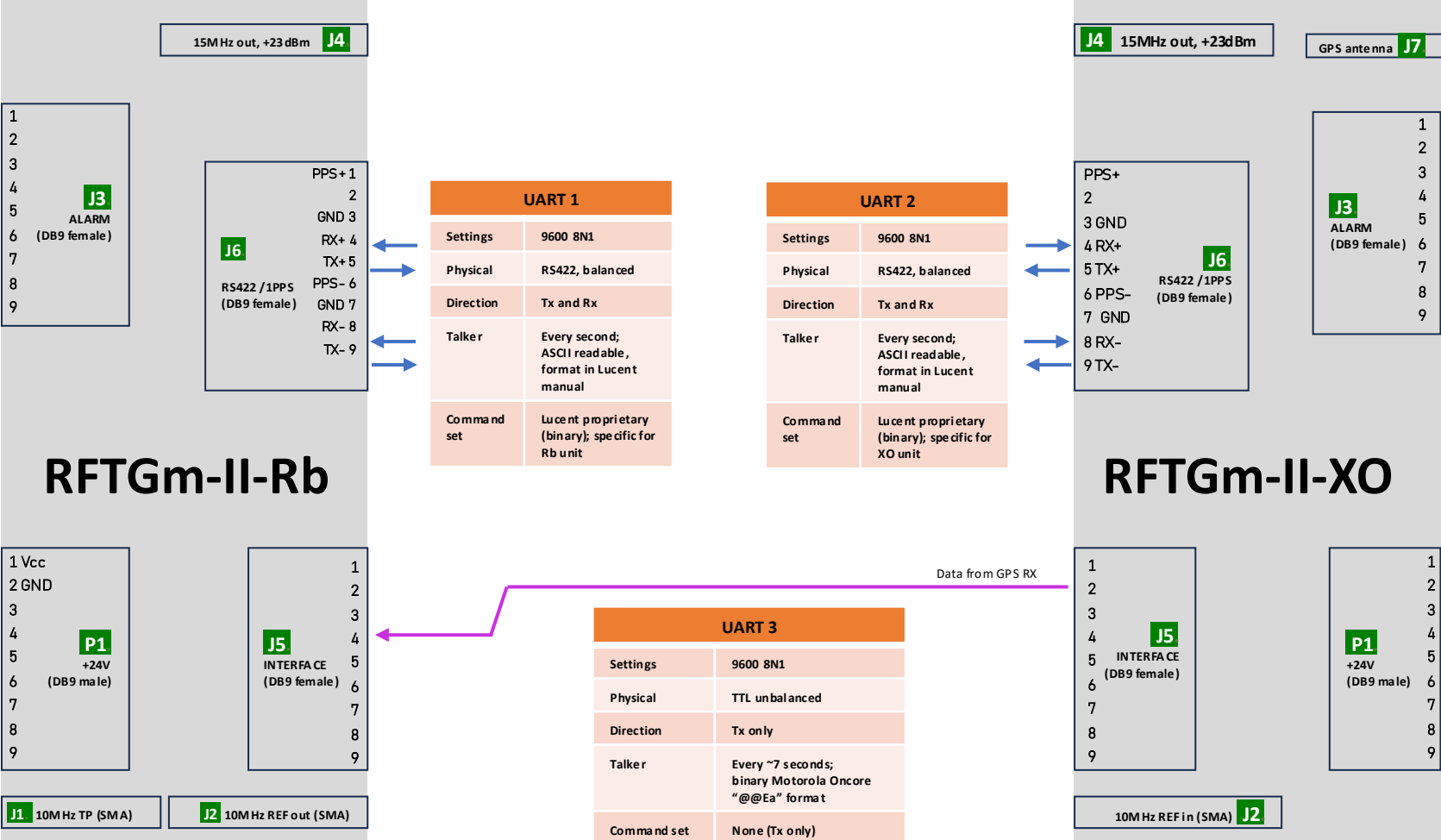
Various RFTG/RFG modifications

Skip Withrow (f22)	Lucent RFTGm-II-XO Modification to Add 10MHz Output.
Charles S. Osborne k4cso (23)	Lucent RFTG-m-XO 15MHz -> 10MHz.
Dan Watson (19)	Dan Watson (Sync Channel Blog). Adding a 10MHz Output to the KS-24361 REF-1. Also: Example Code for KS-24361 REF-0 Standalone Operation .
Perdrix (3)	KS-24019 RFTGm-II-Rb and RFTGm-II-XO, 10MHz output mod. Takes output from Rb and OVXO respectively, and mounts tiny daughterboard on the spot where the 15MHz BPF was located.
Guido Kueppers (9)	Discipline isolated Lucent RFTG-m-RB unit with external supply of PPS and Motorola Oncore "@@Ea" messages.
Tom Knox (34)	Lucent RFTG and RFG 10MHz main output but the full mod is not there!
Gerhard W. Hoffmann (17)	A Frequency Doubler and Distribution Amplifier for the Lucent KS-24361 (RFTG-u) GPS Disciplined Reference Generator . Note: unlike the other RFTG/RFG systems, the RFTG-U has a 5 MHz OCXO, so this mod is not useful for other family members.
Paul Wade W1GH (31)	Doubler: 5 MHz to 10 MHz.
Nigel GM8PZR (32)	Running the RFTG-u REF-1 unit alone. See also « Reply #142 on (33). Apparently only one resistor needed.
John C. Roos, K6IQL (35)	"Converting a Vintage 5 MHz Frequency Standard to 10 MHz with a Low Spurious Frequency Doubler," QEX, March/April 2011, pp. 19-35.
Peter Garde (35)	Standalone GPSDO Operation of the Lucent KS-24361 REF0 (RFTG-u).

RFTGm-II connections (1/3)



RFTGm-II connections (2/3) Serial protocols



Original cable harness

